

Research Article

Improved intelligent water drop-based hybrid feature selection method for microarray data processing

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ABSTRACT

Classifying microarray datasets, which usually contains many noise genes that degrade the performance of classifiers and decrease classification accuracy rate, is a competitive research topic. Feature selection (FS) is one of the most practical ways for finding the most optimal subset of genes that increases classification's accuracy for diagnostic and prognostic prediction of tumor cancer from the microarray datasets. This means that we always need to develop more efficient FS methods, that select only optimal or close-to-optimal subset of features to improve classification performance. In this paper, we propose a hybrid FS method for microarray data processing, that combines an ensemble filter with an Improved Intelligent Water Drop (IIWD) algorithm as a wrapper by adding one of three local search (LS) algorithms: Tabu search (TS), Novel LS algorithm (NLSA), or Hill Climbing (HC) in each iteration from IWD, and using a correlation coefficient filter as a heuristic undesirability (HUD) for next node selection in the original IWD algorithm. The effects of adding three different LS algorithms to the proposed IIWD algorithm have been evaluated through comparing the performance of the proposed ensemble filter-IIWD-based wrapper without adding any LS algorithms named (PHFS-IWD) FS method versus its performance when adding a specific LS algorithm from (TS, NLSA or HC) in FS methods named, (PHFS-IWDTS, PHFS-IWDNLSA, and PHFS-IWDHC), respectively. Naïve Bayes(NB) classifier with five microarray datasets have been deployed for evaluating and comparing the proposed hybrid FS methods. Results show that using LS algorithms in each iteration from the IWD algorithm improves F-score value with an average equal to 5% compared with PHFS-IWD. Also, PHFS-IWDNLSA improves the F-score value with an average of 4.15% over PHFS-IWDTS, and 5.67% over PHFS-IWDHC while PHFS-IWDTS outperformed PHFS-IWDHC with an average of increment equal to 1.6%. On the other hand, the proposed hybrid-based FS methods improve accuracy with an average equal to 8.92% in three out of five datasets and decrease the number of genes with a percentage of 58.5% in all five datasets compared with six of the most recent state-of-the-art FS methods.

1. Introduction

Medical data can be classified into two main categories; first one represents the structured data which is called microarray data, while second category represents an unstructured data that can be two dimensional as in medical imaging, or one dimensional as in biomedical signal processing. Each type of these data always contains some irrelevant or redundant features that must be removed; therefore, the FS process is indispensable for medical datasets analysis for many purposes like diagnosing, screening, and treatment (Remeseiro and Bolon-Canedo, 2019).

Microarray datasets contain many noise features that must be removed for improving classification performance (Li et al., 2017). FS is one of the dimensionality reduction approaches that cover the process of finding the minimal and most optimal subset of features from total features in high dimensional datasets by removing irrelevant and redundant features for increasing the classification or regression accuracy in the supervised learning problems (Mafarja and Mirjalili, 2019).

There are many FS methods developed for finding a more compact and optimal subset of features (Diao and Shen, 2015) in the literature that can be classified based on the availability of supervision like target

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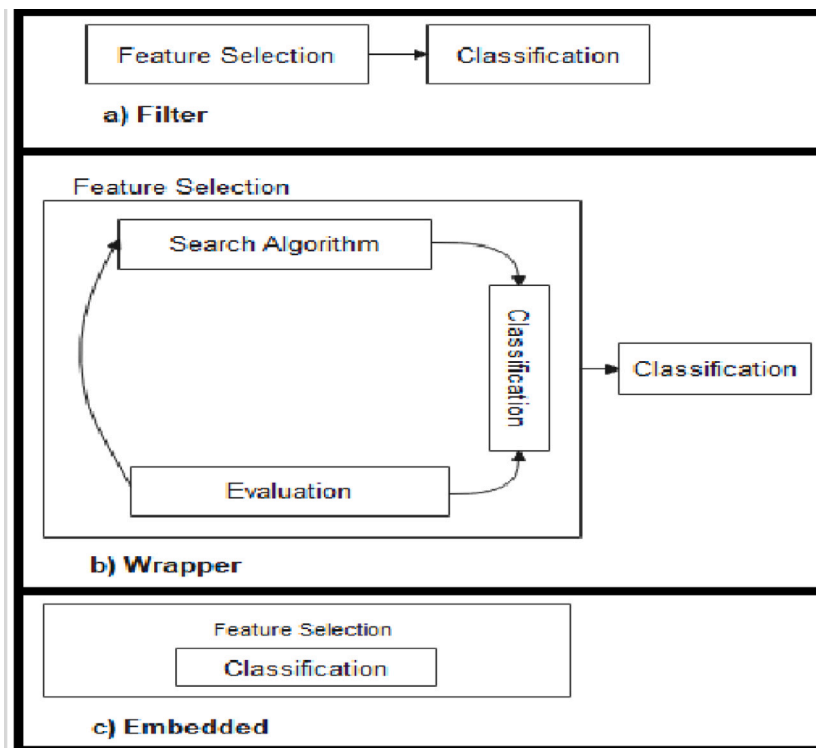


Fig. 1. Some FS methods: (a) Filter, (b) Wrapper, and (c) Embedded.

variable class label into three categories as supervised, semi-supervised or unsupervised FS method, where FS methods in a supervised category were devoted for classification or regression problems based on target variable type as categorical or numerical respectively. In contrast, unsupervised FS methods have usually been used for handling a clustering problems (Li et al., 2017).

Another classification perspective proposed based on the selection strategy as a filter, wrapper, embedded and hybrid feature selection methods, where filter feature selection approach use a statistical technique to provide a rank for each feature then select specific top ranked features based on a specific threshold. In contrast, the wrapper approach works by selecting a subset of features using a specific search algorithm then evaluating it using a specific classifier and fitness function (Li et al., 2017), see Fig. 1 part(a) and part(b).

In an embedded FS methods, the selection accomplished as a part of classification algorithm like in the random forest, where a classifier selects features that maximize accuracy automatically or during the training of the classifier like specifying weights in a neural network (Ang et al., 2015), see Fig. 1 part(c))

On the other hand, hybrid FS methods select subset of features by combining two or more FS methods from various selection strategies for handling their advantages simultaneously, where as in general, the wrapper approach provides more accurate results when compared with the filter approach. However, the filter and embedded approaches need minimum computation time when compared with the wrapper-based FS methods (Manikandan and Abirami, 2018).

In some studies like (Saeys et al., 2008; Abeel et al., 2010; Haury et al., 2011; Manikandan and Abirami, 2018; Rouhi and Nezamabadi-Pour, 2020), authors discussed an additional feature selection strategy and named it an ensemble FS methods, which aims to solve instability and perturbation issues in many individual FS methods by running a particular FS method on several sub-samples, and merging the obtained features to form a more stable subset for tackling an over-fitting problem for high dimensional datasets (Ang et al., 2015).

Most of the microarray datasets in the medical field are high dimensional as it consists of a large number of features compared with a

small number of samples of a particular disease, which may lead to data over-fitting problem and inaccurate results when using wrappers or embedded FS methods because the classifier is repeatedly called to evaluate each subset. Therefore, most traditional FS methods used filter FS methods, especially in microarray data. Recently, there exist a tendency for using hybrid (filter - wrapper) or ensemble methods for FS over medical datasets by applying an ensemble filter at first for dimensionality reduction through removing some features then using a wrapper for fine-tuning by handling a more accurate results (Bolon-Canedo et al., 2015b).

FS methods can be also classified from a data perspective as FS methods with a conventional data, FS with structured features, FS with heterogeneous data, and FS with streaming data (Li et al., 2017).

Feature selection methods with a conventional data include most of the existing FS methods which ignore the inherited feature structures and assume that all features are independent of each other's. This type of FS methods assesses the importance of each feature based on one of four ways: features similarity like "Fisher Score", using various heuristic filter criteria like "Mutual Information", using some sparse regularization terms such as "Multi-Cluster", or using various statistical measures like "Low Variance, and T-score". In contrast to these, some of the FS methods takes the structure of features (spatial or temporal, groups, trees or graphs) into account in FS process therefore, it improves the learning task.

Some FS methods, however, are applicable for heterogeneous data includes data from multiple sources, from multiple views, and linked data such as data in social media. Finally, some FS methods are suitable for streaming features or streaming data, where one dimension; either the number of instances or the number of the candidate features is constant, while the other dimension is unknown or infinite and arrives one at a time like "Unsupervised Streaming" FS method in Social Media. Alhenawi et al. (2022) categorized researches for gene expression classification conducted during the recent seven years based on their purpose into nine directions .

In this paper, we came up with a hybrid feature selection method for microarray data classification, which balance exploration and exploitation capabilities in selecting a subset of features by combining an

Table 1
IWD based FS related work from 2015 to 2020.

Author	Algorithm	Classifier	Application	Datasets	Year
Kalita et al. (2022a)	IWD based FS	SVM.	Breast Cancer detection	MIAS.	2022
Kalita et al. (2022b)	Two-way Threshold Based IWD for FS	SVM.	Breast Cancer detection.	MIAS.	2022
Hariyale et al. (2020)	H-IDS as a hybrid IWD with SVM.	SVM classifier	Intrusion detection system.	KDD Cup 99 dataset.	2020
Singh (2019)	The original IWD.	Naïve Bayes classifier	Email spam classification.	Spam base dataset from the UCI repository.	2019
Parlar and Sarac (2019)	Original IWD compared with ReliefF.	Maximum Entropy	Sentiment analysis.	Turkish Twitter dataset.	2019
Khosravi and Bagherzadeh (2019)	IWD with class separability as an objective function.	Not mentioned.	Feature selection.	Seven benchmark datasets obtained from the UCI repository, where input features vary between 3, and 102.	2019
Alijla et al. (2018)	MRMC-IWD model.	C4.5, VQNN, and SVM.	Feature selection for safety surveillance, fraud detection, clinical management, and healthcare	Seven real-valued UCI FS, and classification benchmark, and two real world datasets called human motion detection, and motor fault detection.	2018
Acharya and Singh (2018)	IWD with SVM.	SVM classifier	Intrusion detection.	KDD CUP'99 dataset.	2018
Haneef and Singh (2017)	IWD with ACO	SVM.	Intrusion detection.	KDD CUP'99 dataset.	2017
Alijla et al. (2015)	MRMC-IWD model.	Not mentioned.	FS.	Thirteen rough-set FS datasets with feature numbers vary between 13, and 56 feature.	2015

ensemble filter method and an Intelligent Water Drop (IWD) algorithm, which was developed by Shah-Hosseini (2009) with two improvements:

- One of the improvements targets IWD exploration capability by using three different iterative Local Search Algorithms (ILSA).
- Another improvement targets the selection process of the next feature added to the IWD solution list for each drop in the original IWD algorithm using correlation coefficient (cc) between current features in IWD list and all other unvested features as a HUD value for updating the soil value carried by the drop itself and soil over the selected path for each IWD agent to eliminate the probability of selection of any redundant features. Fast correlation coefficient filter is a multivariate FS method finds features that are strongly correlated with a specific class and features that have the lowest correlation with other features (Djellali et al., 2017), where among two strongly correlated features; one of them (that has the lowest correlation with specific class) is considered as a redundant feature and not selected as next feature in IWD list.

The main contributions of this paper can be summarized as follows:

1. summarize the most recent state-of-the-art works related to FS methods based on IWD.
2. proposing a hybrid feature selection algorithm for medical applications based on an ensemble filter and an improved IWD as a wrapper.
3. improving an exploitation capability of IWD by adding one of three LS algorithms (TS, NLSA, HC) after each iteration from IWD to eliminate a local optima problem.
4. Evaluating the proposed feature selection algorithm performance against some of the most recent FS algorithms from the literature.

The rest of this paper is organized as follows: Section 2 presents a brief review of the latest works on FS using IWD. Section 3 presents

IWD inspiration and mathematical equations, while Section 4 is devoted to present the proposed hybrid FS method. In Section 5, experimental setups are explained. Experimental results presented and discussed in Section 6. Finally, Section 7 presents the conclusion and future work.

2. IWD based feature selection related work

In the literature, there are some prior work on deploying IWD for FS problem for different applications such intrusion detection, spam email detection, sentiment analysis, web page classification, rough set FS, and gene selection for cancer prediction in the medical field, as illustrated in details in this section and summarized in Tables 1 and 2.

Hendrawan and Murase (2011) developed four embedded feature selection methods to find the most significant set of textual features for an irrigation system. Neural intelligent water drop (N-IWD), which searches for the best subset of features while optimizing weights in a neural network, is one of these methods, which applied over 120 textual features and compared with other three methods called neural simulated annealing, neural-genetic algorithm and neural-discrete particle swarm optimization. Results show that N-IWD outperformed all other three methods.

Alijla et al. (2013) used an Intelligent Water Drops algorithm for Rough Set Feature Selection IWDRSFS as an evaluation function for feature selection. They used six benchmarks, where most of these datasets taken from the UCI machine learning repository with input features vary between 13 and 56.

IWDRSFS was compared with four state-of-the-art rough-set methods for feature selection. It provides better results than local-search-based methods like hill climbing and entropy-based while it is comparable with population-search-based methods like an ant colony, and genetic algorithm.

Alijla et al. (2015) improved an exploration capability for intelligent water drop algorithm by developing an ensemble intelligent water drop using divide and conquer Multiple-Creek IWD (MRMC-IWD) model which an ensemble IWD by using divide and conquer strategy where

Table 2
IWD based FS related work from 2011 to 2014.

Author	Method	Classifier	Application	Datasets	Year
Alijla et al. (2014)	Modified version of IWD using two ranking-based selection methods called linear ranking, and exponential ranking as an IWD fitness proportionate selection method.	Not mentioned	Rough-set feature subset selection, multiple knapsack, and traveling salesman problems	Thirteen benchmark datasets. Most of the datasets are from the UCI repository.	2014
Alijla et al. (2013)	IWDRSFS method.	Not mentioned	FS.	Six datasets obtained from the UCI machine learning repository, where input features vary between 13, and 56 feature.	2013
Kumar et al. (2013)	Weighted gene Hybrid IWD-SVM.	SVM.	Gene selection and cancer prediction.	Three medical datasets were obtained from the Kent Ridge Biomedical datasets repository, and the libSVM repository, where it provides good performance.	2013
Ayse Ozel and Sarac (2012)	IWD with J48.	J48.	Web page classification.	WebKB, and Conference datasets.	2012
Hendrawan and Murase (2011)	Four embedded feature selection methods to find the most significant set of textual features for an irrigation system.	Not mentioned.	Textual features for the irrigation system.	One hundred twenty textual features extracted from grey, RGB, HSV, and HSL.	2011

they divides the original problem into multiple smaller sub-problems then solving each one using specific independent creek finally all solutions from all sub-problems combined to provide the last solution which handled by the master river.

In addition, for improving exploitation capability and enhancing the final solution to be optimal one or closest to the optimal solution, they added an iterative improvement local search (IILS) to the MRMC-IWD model then renamed it as a Hybrid MRMC-IWD model. They applied this model over 13 rough set feature selection (RSFS) datasets with feature numbers varies between 13 and 56 feature where results show that the hybrid MRMC-IWD model outperformed their improved IWD without IILS, and three state-of-the-art methods.

Another modified version of IWD was developed by [Alijla et al. \(2014\)](#), they used two ranking-based selection methods, called linear ranking IRS-IWD, and exponential ranking ERS-IWD, instead of the original IWD fitness proportionate selection (FPS) method for increasing the search diversity and avoid it from trapped into a local optimum.

They evaluated the performance of these proposed models over 13 rough-set feature selection (RSFS) datasets with a number of features vary between 13, and 56 features. Results show that ERS-IWD outperformed LRS-IWD, while FPS was computationally better than ERS-IWD, and LRS-IWD.

An improvement in IWD based feature selection method was done by [Khosravi and Bagherzadeh \(2019\)](#) through using a new objective function suitable for IWD based on class separability strategy by maximizing distance between different classes and minimizing distance within the same class.

They evaluated the proposed algorithm over one artificial dataset and seven benchmark datasets from the UCI repository, where features are varying between 3, and 102. Results show that the proposed approach increases accuracy and decreases the number of features at the same time.

[Alijla et al. \(2018\)](#) applied an ensemble model of the intelligent water drop called the Master River Multiple Creeks IWD (MRMC-IWD) model, which they proposed in 2015 to real-world optimization problems related to feature selection, and classification problems.

They used seven benchmark datasets from the UCI with a number of features vary between 35, and 280 for evaluating their method. Also, two real word datasets called human motion detection, and motor fault

detection used for evaluation with three classifiers called C4.5, VQNN, and SVM. The results present the usefulness of MRMC IWD in tackling real-world FS problems.

[Singh \(2019\)](#) applied the original IWD algorithm as provided by hosini for spam emails classification, where IWD was used for feature construction with NB classifier for classifying emails as spam or not spam. The proposed method evaluated over Spam base dataset from the UCI Repository that contains 4601 observations, and 57 features, where 1813 email are spams. Results show that an IWD-based FS outperformed GA, and ACO.

Another utilization of the original IWD based feature selection was done by [Parlar and Sarac \(2019\)](#) for sentiment analysis over a Turkish Twitter dataset. They compared the proposed FS method performance versus one of a well-known filter-based FS methods called ReliefF in terms of f-score, precision and recall evaluation metrics, where IWD outperformed ReliefF. They used Maximum Entropy as a classifier in their experiments.

Also, [Ayse Ozel and Sarac \(2012\)](#) deployed the original IWD in feature selection for web page classification. They used the J48 classifier for evaluating the fitness for the selected subset of features. The proposed FS system evaluated over WebKB, and Conference datasets. Results present the effectiveness of this system in FS process for web page classification.

[Acharya and Singh \(2018\)](#) developed a FS method based on IWD with SVM for classification in an intrusion detection system. They applied the proposed method over KDD CUP'99 datasets from the UCI machine learning repository, where results illustrated that IWD based FS with SVM provides better performance in terms of accuracy, false alarm rate and detection rate compared with results taken from pure SVM, K -mean, naive Bayes, and GA with SVM.

[Hariyale et al. \(2020\)](#) developed a hybrid intrusion detection strategy that combining IWD with SVM called H-IDS for feature selection, and classification, where H-IDS provides better performance compared with pure K mean and SVM. They used KDD CUP'99 datasets for evaluation.

[Kumar et al. \(2013\)](#) came up with a model for gene subset selection simultaneously with cancer prediction using IWD for gene selection, and SVM for classification. Also, they used a weighted gene ranking value, which composed of three filters, namely Information Gain (IG),

Chi square(CS), and Correlation based feature selection (CFS) for guiding the IWD search. They evaluated the proposed model performance over three medical datasets obtained from the Kent Ridge Biomedical datasets repository, and the libSVM repository, where it provides a good performance.

Haneef and Singh (2017) developed a hybrid FS method based on IWD, and ACO in intrusion detection systems. They used KDD CUP'99 datasets for evaluating the proposed algorithm performance compared to other existing approaches in FS, where it was finding minimal, and most optimal features set with minimal training time. They used SVM as a classifier (see Table 2).

From the literature, it is notable that almost all studies that employed an IWD for FS, documented a superior or sometimes comparable results for the proposed IWD-based embedded or wrapper FS methods compared with methods like GA, PSO, ACO and others. Most of these studies focused on specific applications like intrusion detection, spam email detection, sentiment analysis, and web page classification.

Also, we found some studies deployed IWD based FS for cancer prediction like (Kumar et al., 2013) study, where authors used the original IWD with a new way for guiding an IWD search process in gene subset selection for cancer prediction, and results illustrated shown to provide good performance in terms of the number of selected genes and classification accuracy compared with some of a FS methods used for the same application.

Kalita et al. (2022a,b) are two of the most recent studies that deployed IWD based FS for Breast Cancer prediction, where both works design a computer aided diagnosis system by extracting specific features in wavelet domain from mammograms in the first stage. In the next stage (Kalita et al., 2022a) used an IWD algorithm to select the most optimal set of features from the extracted features set while (Kalita et al., 2022b) used a two way threshold based IWD FS algorithm for Breast cancer prediction in the early stages. SVM used in both papers for classification with a Mammographic Image Analysis Society (MIAS) database to detect the efficiency of the proposed technique versus other meta heuristic FS techniques like Particle Swarm Optimization (PSO), Genetic Algorithm (GA), and Ant Colony Optimization (ACO). Results show that the proposed technique outperformed all other algorithms and achieving an accuracy of 99%, 97.9% in Kalita et al. (2022b,a), respectively.

In this paper, however, we aim to improve microarray data classification performance using as little as possible number of genes, by using ensemble filter-IIWD-based wrapper that employed one of three common LS algorithms at a time for increasing an IWD exploitation capability compared with it without any LS algorithms.

3. IWD algorithm

IWD was developed by Shah-Hosseini (2009). It is inspired the way by which water drops flow in natural rivers intelligently, where each drop work as an independent agent that was initially starting from the source and moves randomly with specific velocity and specific initial value of soil. During movement, each drop will carry an amount of soil from the bed of the path proportional to its velocity. The soil in the path will be decreased as the shortest and best path to encourage other drops to follow it because water drops always prefer to follow the bath with less soil (Shah-Hosseini, 2012).

General steps and mathematical equations in IWD algorithm based on Shah-Hosseini (2009) are as follows:

1. For solving any problem using IWD algorithm, it must be translated into a graph $G(N, E)$ with nodes N and edges E that connected nodes, where each edge has some amount of soil called $init^{soil}$, which is a user-selected value tuned experimentally based on an application. Soil between any two nodes i, j set initially equal to $init^{soil}$, as shown in Eq. (1).

$$Soil(i, j) = init^{soil} \quad (1)$$

2. Initialization of static parameters like

- * Quality of total best solution $q(T^{TB})$, which set initially equal to the worst value and it differ from one problem to another.
- * The total number of iterations $iter_{max}$, which is specified by users, and counted for current iteration number $iter_{count}$, which initially sets to zero.
- * The total number of IWD agents IWD_{max_num} , which is a positive number that usually equal to the number of nodes in the graph.
- * Static parameters for soil updating a_s, b_s , and c_s , where all set equal to one, and velocity updating parameters a_v, b_v , and c_v , which are set equal to one.
- * Global soil updating parameter $\rho^{IWD_{num}}$, which selects from $[0, 1]$, and local soil updating parameter ρ^n , which is a small positive number less than one.

3. Initialization of dynamic parameters like

- * Every IWD has a specific velocity $vel^{IWD}(t)$, which updates later at each movement for IWD based on Eq. (3). $vel^{IWD}(t)$ initially = $Init^{Vel}$, which is a user-selected value, and it should be tuned experimentally based on an application.
- * Every IWD has specific soil $soil^{IWD}$, which initially sets equal to zero, then it updates based on Eq. (6).
- * Every node has a list of visited nodes, which is initially empty $vc(IWD)$.

4. Spread the predefined number of IWDs that represent agents randomly on the nodes, where each agent will have started from a specific node as its first visited node in its solution list.
5. Steps from 6 to 7.3 repeated for those IWDs with partial solutions until reaching a specific iteration-best solution I^{BS}
6. Each IWD transmits from current node i to the next unvisited node j that does not violate some constraints specified in the target problem based on Eq. (2), which depends on soil value between the current node and other nodes, where drops prefer transmission through the fewer soil paths. Through transmissions from one node to another, each drop stores the list of visited nodes for determining its solution path.

$$p_i^{IWD}(j) = \frac{f(soil(i, j))}{\sum_{k \notin vc(IWD)} (f(soil(i, k)))} \quad (2)$$

such that

$$f(soil(i, j)) = \frac{1}{\epsilon s + g(soil(i, j))}$$

$$g(soil(i, j))$$

$$= \begin{cases} soil(i, j) & \text{if } minsoil_{IWD}(i, l) \leq 0 \\ soil(i, j) - minsoil_{IWD}(i, l) & \text{else} \end{cases}$$

7. After each drop transition, updating the dynamic parameters
 - 7.1. Update the velocity of each IWD moved from i to j $vel^{IWD}(t)$ based on Eq. (3)

$$vel^{IWD}(t+1) = vel^{IWD}(t) + \frac{a_v}{b_v + c_v \cdot soil^2(i, j)} \quad (3)$$

Where,

$vel^{IWD}(t+1)$ is the updated velocity for the water drop.

- 7.2. Compute the amount of soil that IWD loads through its movement over the path from i to j ($\Delta soil(i, j)$), which depends on the HUD(j) that defined appropriately for the given problem as shown in Eq. (4).

$$\Delta soil(i, j) = \frac{a_s}{b_s + c_s \cdot time^2(i, j; vel^{IWD}(t+1))} \quad (4)$$

Where,

$$time^2(i, j; vel^{IWD}(t+1)) = \frac{HUD(j)}{vel^{IWD}(t+1)}$$

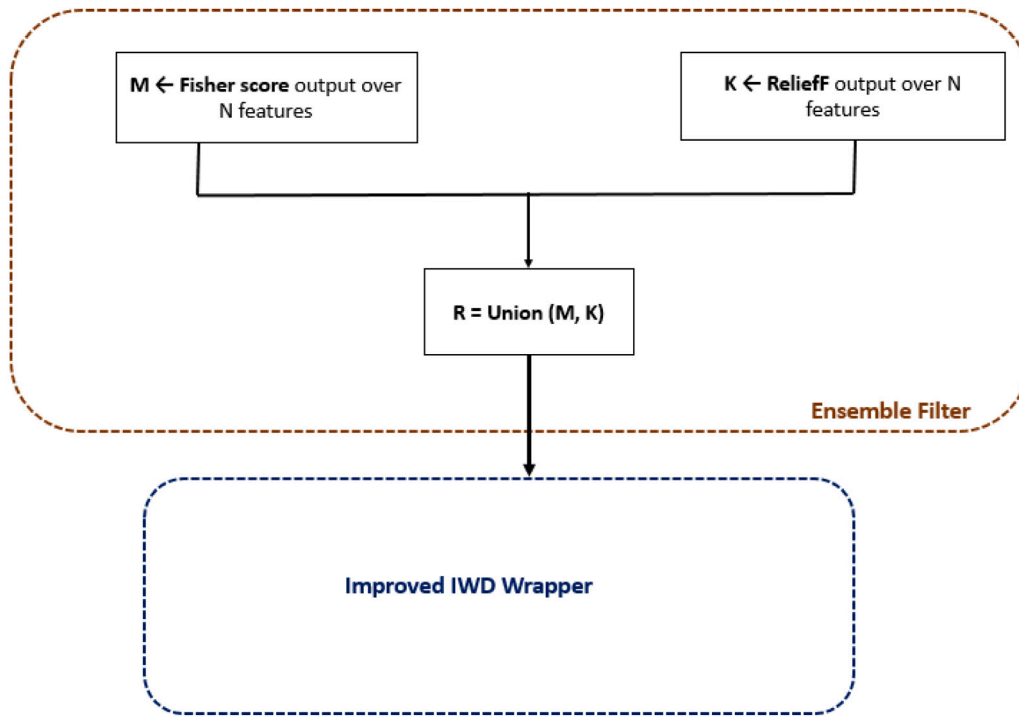


Fig. 2. The proposed hybrid FS algorithm.

7.3. Update the soil of the path between i and j as $(soil(i, j))$ and the soil carried by IWD itself as $soil^{IWD}$ based on Eqs. (5) and (6), respectively.

$$soil(i, j) = (1 - P_n) \cdot soil(i, j) - P_n \cdot \Delta soil(i, j) \quad (5)$$

$$soil^{IWD} = soil^{IWD} \cdot \Delta soil(i, j) \quad (6)$$

8. At the end of each iteration best path, which provides a better fitness value, store as the iteration best solution path T^{IB} found so far by all IWDs in the current iteration, as in Eq. (7).

$$T^{IB} = \arg \max_{T^{IWD}} q(T^{IWD}) \quad (7)$$

Where, $q()$ is a fitness function for the solution, which differs from one problem to another.

9. Update the soil over the selected paths that form the current iteration-best solution T^{IB} , as in Eq. (8).

$$soil(i, j) = (1 + P_{IWD}) \cdot soil(i, j) \quad (8)$$

$$soil(i, j) = (1 + P_{IWD}) \cdot soil(i, j) - P_{IWD} \cdot \frac{1}{N^{IB} - 1} \cdot soil_{IB}^{IWD} \quad \forall (i, j) \in T^{IB}$$

Where, N^{IB} is the number of nodes in T^{IB} .

10. At each iteration, T^{IB} compared with the total best solution T^{TB} ; if it is better, then T^{TB} replaced with it, as in Eq. (9)

$$T^{TB} = \begin{cases} T^{TB} & \text{if } q(T^{TB}) \geq q(T^{IB}) \\ T^{IB} & \text{Otherwise} \end{cases} \quad (9)$$

11. Repeat steps from 3 to 10 until reaching a maximum number of iterations $iter_{max}$, which is specified as a static parameter in step2.
12. Return T^{TB} as the last best solution path for the target problem.

4. The proposed hybrid FS method

In most of the works that developed hybrid FS methods, authors select specific filter for reducing the number of features passed to the

wrapper stage later, where that wrapper must select features from the filtered subset for optimizing the accuracy of a particular classifier on the training set, but this way (means using a single filter) depends only on one filter for selecting features that are passed to the wrapper, which means that the probability of removing some relevant features before providing it any chance to be considered in a wrapper increases, while using multiple filters rather than single one decrease the prior probability and consequently enhances the wrapper final results (Leung and Hung, 2008).

Filter methods can be classified as a ranking-based group and search-space-based group, where in the first group weigh of features specified according to the extended relationships that obtained based on the correlation between all features and label classes then features with weights greater than a specific threshold will be selected. At the same time, other features will be omitted, therefore this type of filters are succeeded in removing irrelevant features but not suitable for removing the redundant features (Lazar et al., 2012).

On the other hand, the second filter group of FS methods uses some assessment functions and other criteria for finding correlation within features for detecting the redundant features and removing them later.

Search space methods have more efficiency than ranking algorithms because it assesses the importance of each feature by measuring correlation between feature and label of classes in addition to correlation within features. This kind of filter algorithm tends to be temporally expensive compared with ranking-based filter group methods as it employed more computations (Aghdam et al., 2009).

In the proposed hybrid FS method, we decrease the overfitting problem risks for the proposed improved IWD based wrapper by applying an ensemble filter initially over the high dimensional microarray datasets.

4.1. Filter part

In this paper, we used an ensemble filter that combines two of the most common ranking-based filters called ReliefF, and Fisher score, then the subsets of features for these filters aggregated using union process. An ensemble filter is used for eliminating irrelevant features and reducing the number of features passed to the wrapper stage (look at Fig. 2).

4.1.1. ReliefF

It is one of the Relief family algorithms that dealing with multi class classification problems. It measures the importance for each feature represented by a specific weight depending on how well their values distinguish between instances that are near to each other.

The process is done by selecting random instance each time and ReliefF searches for k nearest neighbors of it from the same class called “near hit”, and others k nearest neighbors from a different class called “near miss”, then based on the calculated values for “near hit”, and “near miss” the weight value, which is the average for all hits, and misses computed for each feature by repeating this task for (T) times.

This method has many advantages, as it can handle any type of data, and it is not affected by the interaction between features. On the other hand, it cannot find the redundant features (Zhang et al., 2003).

4.1.2. Fisher score

It is one of the most common methods for supervised learning FS. It provides rank for each feature by computing the discriminative power for each feature through fisher score where features that have higher fisher score provides a more relevant distinction between classes which consequently means that this feature has a high priority to be selected for classification (Fu et al., 2019).

4.2. An improved IWD-based wrapper part

For increasing the exploitation capability of the IWD algorithm, and decreasing the risks of the local optima problem, we proposed an improvement in IWD, where in the improved version of IWD, we added one of three different LS algorithms, and compared the performance of NB classifier in the test set when using final subset of features that we obtained from applying the proposed improved IWD wrapper stage with a specific LS algorithm each time over the filtered set of features (R) that comes from the first stage in the proposed hybrid FS method based on F-score and number of selected features.

Also, as another improvement in IWD, a fast correlation coefficient filter was used as an HUD value for guiding IWD search and eliminating selection of redundant features when select the next feature in each IWD solution list.

HUD proportional to an amount of soil which IWD loads through its movement over a path from current node i to the next node j $\Delta soil(i, j)$ which affected on IWD current soil and soil(i, j).

The relationship between HUD(i, j) and the probability of selecting feature j can be represented as follows:

$$HUD(i, j) \propto \frac{1}{\text{probability of selecting the feature}(j)}$$

Therefore, when feature j highly correlated with feature i , HUD will be high; which means $\Delta soil(i, j)$ will be small based on Eq. (4) and consequently soil(i, j) will be small, then the probability for selecting j as the next feature in IWD list will be decreased, where it highly correlated with i (in other words i and j are redundant).

The steps of the proposed hybrid FS method for microarray datasets, as shown in Fig. 3

1. Apply the ensemble filter stage over a microarray dataset that we selected for decreasing number of features that will continue in wrapper stage for improving accuracy results.
2. Translate the dataset as a fully interconnected graph $G(R, E)$, where filtered features from the previous filtering stage (R) represents nodes and transmissions from one feature to another represents edges E .
3. Define the static parameters like

- The number of IWDs (IWD_{max_num}), that defined to be equal to the number of features that returned from filter part(R).
- The total number of iterations ($iter_{max}$), which we tuned experimentally for getting better results.

- Initial soil and initial velocity will be equal for all drops at the beginning, and we tuned them experimentally for getting better results.
- Quality of total best solution $q(T^{TB})$, which equal to F-score as fitness function here.
- Static parameters for soil updating a_s, b_s , and c_s , where all set equal to one and velocity updating parameters a_v, b_v and c_v , which also set equal to one.
- Global soil updating parameter ρ^{IWD} , and local soil updating parameter ρ^n .

4. Initialization of dynamic parameters like

- $Init^{Vel}$ for each IWD, which tuned experimentally for getting the best results.
- $soil^{IWD}$ for each IWD, which is initially equal to zero, and updates through drop movements.
- Every water-drop has a list of visited nodes, which is initially empty $vc(IWD)$.

5. Spread the predefined IWDs that represent agents randomly on nodes that represent features, where each agent finds a specific solution (subset of features).
6. Each IWD (IWD_{num}) transmits to the next unvisited node j , which adds as a new feature to the drop selected features list (IWD solution) based on Eq. (2).
7. Velocity and soil value for each IWD updates after each transition based on Eq. (3), Eq. (5), and Eq. (6), respectively. IWD velocity depends on soil value between the current node, and the selected one, while IWD soil, and selected edge soil values depend on IWD current velocity and HUD values based on Eq. (4), Eq. (5) and Eq. (6).
8. After each IWD completes its solution that contains the predefined number of features ($n_{features}$) [means after a current iteration completes] the solution that provides best F-score value store as the iteration best solution T^{IB} . This solution that found so far by all IWDs in the current iteration computed based on Eq. (7)
9. Update the soil through the selected best path based on Eq. (8)
10. Choose one of the three local search algorithms, that we selected for increasing the exploitation capability of IWD-based wrapper FS algorithm and apply it over T^{IB} , which handle the current iteration best solution then if the solution of the used local search algorithm (LS solution) provides better F-score value, store it in T^{IB} . The total iterations best solution T^{TB} , which stores a final result, will be updated after each iteration with a better solution if it is found or still unchanged based on Eq. (9).
11. Steps from 4 to 10 repeated at each iteration until reaching the maximum number of iterations ($Iter_{max}$).
12. Return the final best subset of features T^{TB} .

4.3. Local search algorithms (LS)

Local search algorithms represent a strategy of search in space by moving from a specific solution to another one in its neighborhood depending on specific rules (Pirlot, 1996). In the proposed improved IWD wrapper, we deployed three local search algorithms which listed and displayed in this section.

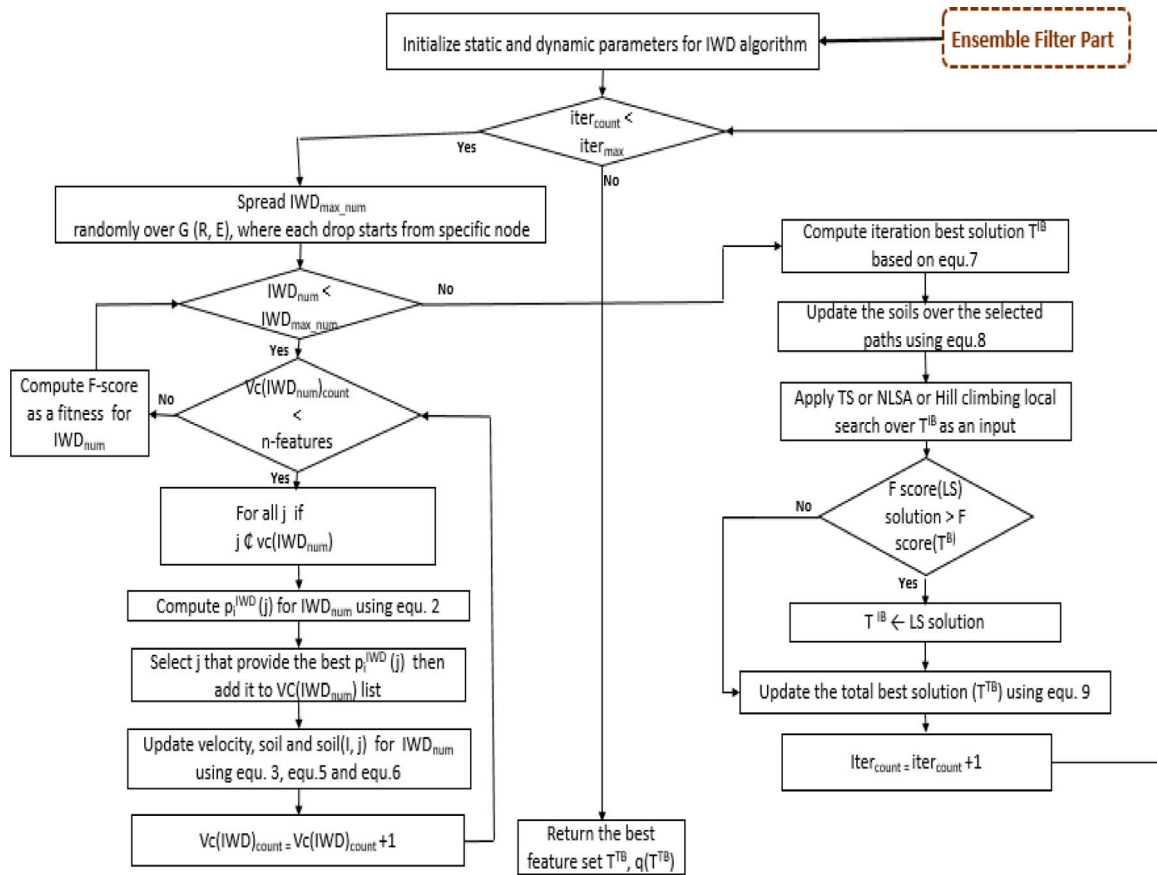


Fig. 3. The proposed Improved IWD-based wrapper FS algorithm.

4.3.1. Tabu search (TS)

Tabu search proposed by Glover (1997). It is a local search technique guided by using an adaptive memory structure called Tabu list, which stores each visited point for avoiding entrapment in local optima (Henderson et al., 2003).

This method starts with a single solution selected randomly in general (in our proposed work the first solution will not be chosen randomly. Instead, it will be the last iteration best solution returned from each IWD iteration T^{IB}) where this solution will be updated to one of its current neighbors that provides better results. This process is repeated until satisfying the specific constraints (Alazzam et al., 2019).

4.3.2. Novel local search algorithm (NLSA)

NLSA starts with T^{IB} solution produced by IWD at the end of each iteration. It iterates for specific times to improve the current solution by selecting three features randomly from it; then NLSA flips the value of the selected features for generating a new solution, and updates T^{IB} with the new solution if the last one provides a better F-score. Otherwise, the current T^{IB} remains unchanged (Tubishat et al., 2020).

4.3.3. Hill climbing (HC)

It is an iterative mathematical optimization technique that starts from a specific solution and continues updating it with a better solution in the neighborhood if it is found through evaluating the current solution (Shehab et al., 2017).

The proposed hybrid FS method enhances the final solution to be optimal one or closest to the optimal solution by employing an IWD-based wrapper for FS, which employed some of the common LS algorithms to improve the exploitation capability of IWD and prevent an overfitting problems or trap into a local optima problem. Also, this method supports robustness and stability of results using an ensemble filter in the first stage.

Table 3

Summarize characteristics of microarray datasets that were used.

Dataset	#features	#samples	#classes
Colon Tumor	2000	62	2
DLBCL-Stanford	4026	47	2
CNS	7129	60	2
Prostate cancer	12600	136	2
Ovarian cancer	15154	253	2

5. Experimental setups

This section devoted for displaying the main setups for experiments that have been conducted in this paper including information's about datasets, data preprocessing steps, parameters tuning, classifier, and evaluation metrics.

5.1. Datasets

Five benchmark microarray datasets have been used in this work called Colon Tumor, diffuse large B-cell lymphoma(DLBCL)-Stanford, Central Nervous System(CNS), Prostate cancer, and Ovarian Cancer (NCI PBSII Data) from ELVIRA Biomedical Data Set Repository. Characteristics of these datasets summarized in Table 3.

All these datasets have been divided randomly into training set that contains two-thirds of the samples, and testing set that handles the remaining third samples.

5.1.1. Colon tumor:

contains 2000 genes and 62 samples, where 40 sample labeled as "negative" taken from tumor parts of some patients against 22 samples

Table 4
Static parameters values.

Parameter	Value	Description
a_v	1000	For velocity updating
b_v	0.01	For velocity updating
c_v	1	For velocity updating
a_s	1000	For soil updating
b_s	0.01	For soil updating
c_s	1	For soil updating
$iter_{max}$	1000	Maximum number of iterations
$soil^{IWD}$	10000	The initial soil on each path (edge)
I_{init}^{Vel}	200	The initial velocity of each IWD
ρ^{IWD}	0.9	The global soil updating parameter
ρ^n	0.9	The local soil updating parameter
$IWD_{max,um}$	Number of genes	Number of drops are same as number of genes in the input dataset

labeled as “positive” of normal biopsies taken from other parts of the same patients.

5.1.2. DLBCL-Stanford:

contains 47 samples of different types of DLBCL using a gene expression data, where twenty-four of samples are from “germinal centre B-like” type versus twenty-three from “activated B-like” type. Sample is described by 4026 genes.

5.1.3. CNS:

contains 7129 genes with 60 samples for analyzing the outcome of the treatment provided to the patients of a CNS embryonal tumor. Twenty-one of samples are for patients who responded to the treatment “Class1”, and thirty-nine of samples for patients who does not responded to the treatment “Class0”.

5.1.4. Prostate cancer:

contains 12600 genes versus 136 samples. Fifty-nine of samples labeled as a “Normal”, and 77 as a “Tumor”.

5.1.5. Ovarian cancer(NCI PBSII data):

contains 15154 genes with 253 samples, where 91sample labeled as a “Normal”, and 162 labeled as “Cancer” for predicting ovarian cancer cases from a normal cases.

5.2. Data preprocessing

We translated class column inputs to binary classes 0 or 1 in all datasets. Also, the data in the five microarray datasets were normalized to zero mean and 1 variance to prevent a feature with large scale values from dominating either the selection or classification processes and obtained more accurate classification results.

5.3. Parameters tuning

In all experiments, some parameter values for IWD and LS algorithms are set depending on trial and error, while others tuned based on recommendation of the previous studies. Table 4 presents some of these parameters.

5.4. Classifier

In our work, we used NB as a classifier for training the model based on the selected subset of features obtained using the proposed FS method and then evaluated the trained model over the testing set.

NB classifier is a simple classification algorithm based on Bayes theorem with independence and normality assumptions among the variables (Friedman et al., 1997). NB is one of the most popular classifiers for class prediction or pattern recognition from microarray gene expression data (Ahmed et al., 2017).

We used simple NB as a classifier while SVM (Kalita et al., 2021), Artificial Neural Network, or any other sophisticated classifier was not explored in our study because we focus on the effects of the proposed hybrid FS method in classification results.

5.5. Evaluation metrics

This section presents the evaluation metrics that we used for evaluated the proposed FS method. We used two evaluation metrics called accuracy and F-score (Shukla et al., 2018)

1. Accuracy:

This performance metric computes the total number of samples that are classified correctly using the proposed algorithm from all classified samples as in Eq. (10).

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (10)$$

2. F-score:

This metric is a harmonic mean of precision and recall as in Eq. (11).

$$F - score = \frac{(2TP)}{(2TP + FP + FN)} \quad (11)$$

All metrics depends on four parameters (Turgut et al., 2018)

- **True Positive:** it represents a number of samples classified correctly as “Class1” according to the algorithm.
- **True Negative:** it represents a number of people who are correctly categorized as “Class0” according to the algorithm.
- **False Positive:** it represents a number of people who are wrongly categorized as “Class1” according to the algorithm.
- **False Negative:** it represents a number of people who are wrongly categorized as “Class0” according to the algorithm.

5.6. Experiments

We conducted experiments in two stages, as shown in this section.

5.6.1. The first stage of experiments

In the first stage, we evaluated the effect of each local search algorithm on IWD performance as a wrapper based FS method by conducting four experiments over each one of the selected microarray datasets and comparing their results based on F-score, where in the first experiment, we adopted the proposed hybrid FS based on IWD without LS algorithm (PHFS-IWD). In contrast, the second experiment conducted using a proposed hybrid FS method based on IWD with Tabu search as an LS algorithm (PHFS-IWDTS).

In the third experiment we used NLSA as a LS algorithm, which was developed by Tubishat et al. (2020) named (PHFS-IWDNLSA). Finally, the fourth experiment was handled using hill-climbing LS algorithm (PHFS-IWDHC).

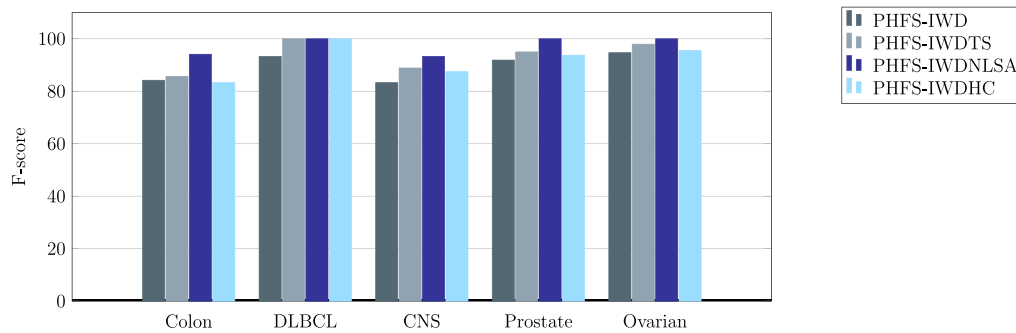


Fig. 4. Comparing the proposed PHFS-IWD FS methods.

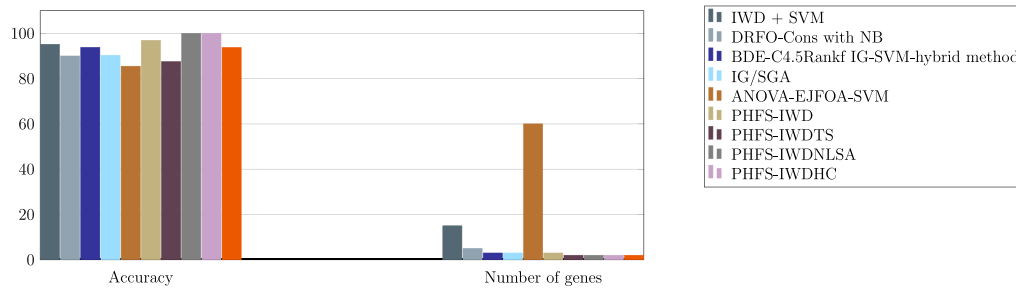


Fig. 5. Comparison of the proposed methods versus some methods in the literature for Colon Tumor dataset.

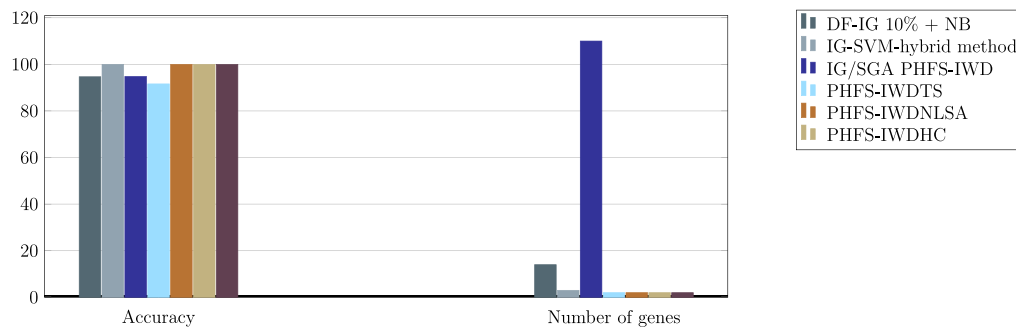


Fig. 6. Comparison of the proposed methods versus some methods in the literature for DLBCL-Stanford dataset.

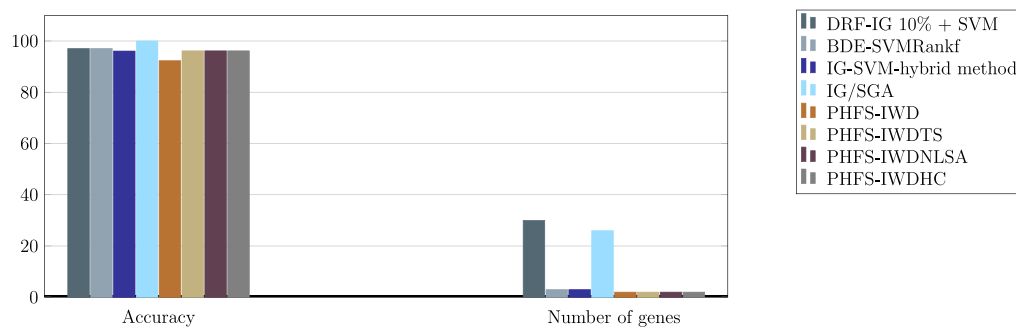


Fig. 7. Comparison of the proposed methods versus some methods in the literature for Prostate dataset.

5.6.2. The second stage of experiments

In this stage of experiments, we provide an approximated comparison of the proposed hybrid FS methods results with six of the most recent state-of-the-art FS methods based on the number of features, and classification accuracy metrics.

Comparison here provides an approximated measurement due to the heterogeneity of the mentioned FS methods in terms of classifiers used, and samples used in the validation process, where some studies

validate their results using cross-validation, others employ a testing dataset for validation. For example, Salem et al. (2017) utilized the 10-fold cross-validation for evaluating the selected genes selected in the training set, while (Apolloni et al., 2016) used the training set to select the features, and the test set to validate the final feature subset. On the other hand, we cannot have implemented all FS methods that involved in the comparison. Therefore, we compared the performance of the best found values by the proposed hybrid FS methods versus

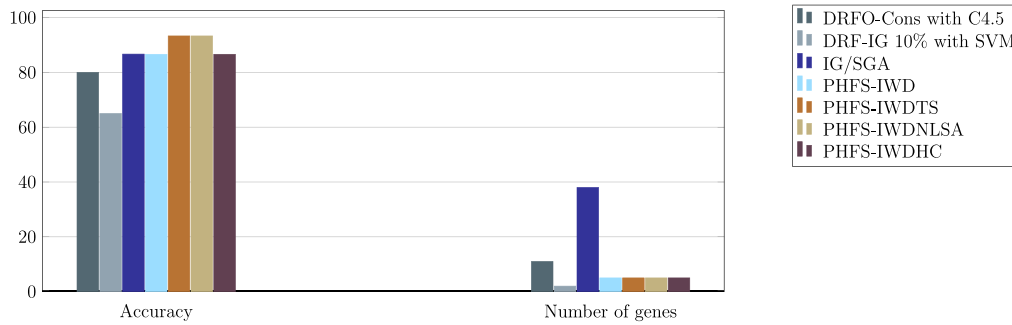


Fig. 8. Comparison of the proposed methods versus some methods in the literature for the CNS dataset.

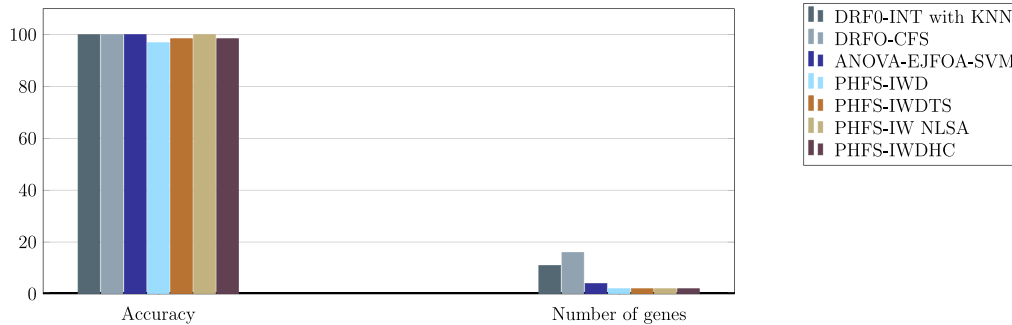


Fig. 9. Comparison of the proposed methods versus some methods in the literature for the Ovarian dataset.

the best results reported in the respective papers for the six selected proposals for comparison in this manuscript.

6. Experimental results and discussion

In this section, we present and discuss the results that we obtained from experiments, that we conducted in two stages.

6.1. Results for the first stage of experiments

This part presents the comparison results based on F-score of applying the proposed hybrid FS method with improved IWD using TS, NLSA, and HC local search algorithms named (PHFS-IWDTS, PHFS-IWDNLSA, and PHFS-IWDHC) respectively versus a hybrid method using an original IWD as a wrapper without any local search algorithms (PHFS-IWD) using NB classifier and five features over five microarray datasets.

From Table 6 and Fig. 4, we observed that PHFS-IWDNLSA outperforms all other methods, as it increases the F-score value with an average of 5.67% over PHFS-IWDHC, 4.15% over PHFS-IWDTS, and 8.27% over the PHFS-IWD method. The PHFS-IWDTS method outperforms the PHFS-IWD method with an average of 4.26%, and the PHFS-IWDHC method with an average equal to 1.6%. On the other hand, the PHFS-IWDHC method provides better results than the PHFS-IWD method in four out of the five datasets, with an average of 2.67%.

From the statistical analysis as illustrated in Table 5 Based on a significance threshold of 0.05, the obtained p-values indicate that the final outcomes were statistically significant.

6.2. Results for the second stage of experiments

In this stage, we discussed the results obtained from comparing the proposed hybrid FS method against six of the most recent FS methods reported in the literature for each dataset, as displayed in tables from Table 7 to Table 11, where each table show results of a specific dataset.

Table 5

Statistical analysis results.

Comparing Methods	P-value
PHFS-IWDNLSA vs. PHFS-IWD	0.000926
PHFS-IWDTS vs. PHFS-IWD	0.012529
PHFS-IWDNLSA vs. PHFS-IWDHC	0.048977
PHFS-IWDNLSA vs. PHFS-IWDTS	0.034839
PHFS-IWDTS vs. PHFS-IWDHC	0.028271

The proposed approach reduces the size of selected genes with an average percentage equal to 57.2% and provides an accuracy reach to 100% using TS, and NLSA local search algorithms for the Colon Tumor dataset.

The PHFS-IWDTS, PHFS-IWDNLSA, and PHFS-IWDHC methods improve accuracy with an average of 8%, 8%, and 2% for the three methods respectively, compared with previous works. (look at Fig. 5).

The proposed approach reduces the size of selected genes with an average percentage equal to 72.4% and provides an accuracy reach to 100% using TS, NLSA, and HC local search algorithms for the DLBCL-Stanford dataset.

The PHFS-IWDTS, PHFS-IWDNLSA, and PHFS-IWDHC methods increase accuracy with average percentages equal to 2.6% for all three methods, compared with four of the most recent FS methods that reported in the literature (look at Fig. 6).

The proposed hybrid FS methods provides competitive results based on the number of selected genes by decreasing the number of selected genes with an average percentage of 63% without any significant impact on the classification accuracy compared with four of the most recent FS methods proposed in the literature for a Prostate Cancer dataset, as shown in Table 9, and Fig. 7.

It is evident in Table 10 and Fig. 8 that PHFS-IWDTS, and PHFS-IWDNLSA methods outperformed the rest two methods with accuracy of 93.3% and accuracy improvement average percentage equal to 16.2% compared with three state-of-the-art FS methods. The proposed hybrid FS methods reduces the number of selected genes with an average percentage of 27.1%.

Table 6

Comparison results based on F-score for all datasets using the Proposed hybrid FS method using improved IWD with different LS algorithms versus the Proposed hybrid FS method using an improved IWD without any LS (PHFS-IWD).

Dataset	PHFS-IWD	PHFS-IWDTS	PHFS-IWDNLSA	PHFS-IWDHC
Colon	84.21	85.7	94.11	83.3
DLBCL-Stanford	93.3	100	100	100
Central Nervous System	83.3	88.88	93.3	87.5
Prostate cancer	91.89	95	100	93.75
Ovarian cancer	94.7	97.95	100	95.6

Table 7

The proposed hybrid FS method versus some hybrid FS methods in the literature using NB classifier based on accuracy and number of selected genes over Colon dataset.

Reference	Method	Accuracy	#Genes	
Kumar et al. (2013)	IWD + SVM	95.16	15	
Bolón-Canedo et al. (2015a)	DRFO-Cons with NB	90	5	
Apolloni et al. (2016)	BDE-C4.5Rankf	93.8	3	
Gao et al. (2017)	IG-SVM-hybrid method	90.32	3	
Salem et al. (2017)	IG/SGA	85.48	60	
Baliarsingh et al. (2020)	ANOVA-EJFOA-SVM	96.90	3	
Our Proposed methods				Features-list
	PHFS-IWD+NB	87.5	2	[302 810]
	PHFS-IWDTS+NB	100	2	[568 693]
	PHFS-IWDNLSA+NB	100	2	[865 267]
	PHFS-IWDHC+NB	93.75	2	[960 970]

Table 8

The proposed hybrid FS method versus some hybrid FS methods in the literature using NB classifier based on accuracy and number of genes over DLBCL-Stanford dataset.

Reference	Method	Accuracy	#Genes	
Bolón-Canedo et al. (2015a)	DF-IG 10% + NB	94.67	14	
Gao et al. (2017)	IG-SVM-hybrid method	100	3	
Salem et al. (2017)	IG/SGA	94.80	110	
Our Proposed methods				Features-list
	PHFS-IWD+NB	91.6	2	[1682 1775]
	PHFS-IWDTS+NB	100	2	[1730 769]
	PHFS-IWDNLSA+NB	100	2	[1184 577]
	PHFS-IWDHC+NB	100	2	[1410 244]

Table 9

The proposed hybrid FS method versus some hybrid FS methods in the literature using NB classifier based on accuracy and number of genes over Prostate Cancer dataset.

Reference	Method	Accuracy	#Genes	
Bolón-Canedo et al. (2015a)	DRF-IG 10% + SVM	97.06	30	
Apolloni et al. (2016)	BDE-SVMRankf	97.1	3	
Gao et al. (2017)	IG-SVM-hybrid method	96.08	3	
Salem et al. (2017)	IG/SGA	100	26	
Our Proposed methods				Features-list
	PHFS-IWD+NB	92.3	2	[6124 4699]
	PHFS-IWDTS+NB	96.15	2	[3681 1796]
	PHFS-IWDNLSA+NB	96.15	2	[2510 5227]
	PHFS-IWDHC+NB	96.15	2	[1481 620]

PHFS-IWDNLSA outperforms all other proposed methods in terms of classification accuracy with an accuracy of 100%, and fewer genes (by up to 73%) less than the number of genes in three of the most recent state-of-the-art FS methods for the Ovarian dataset. (look at Fig. 9, and Table 11)

The proposed hybrid-based FS method is aimed to improve classification performance for microarray data using as little as possible number of genes. We were able to show that the proposed method truly satisfied these purposes as they improved classification accuracy with an average equal to 8.92% in three out of five microarray datasets and reduced the number of selected genes with percentage of 58.5% in all the five datasets compared with six of the most recent state-of-the-art FS methods (see Table 8).

7. Conclusion and future work

In this paper, a hybrid FS method based on an ensemble filter, and an improved Intelligent water drop algorithm (IWD) as a wrapper is proposed. Initially, an ensemble filter used for decreasing the number of features that later passed to the next stage where an improved IWD-based wrapper FS method is applied. Improvement in the original IWD done through two steps; First, a correlation coefficient filter used as the HUD to select the next feature in each iteration, then one LS algorithm from (TS, NLSA, and HC) is used to increase the exploitation capabilities of IWD and eliminate local optima problem.

Five benchmark microarray datasets with NB classifier have been used for evaluating the efficiency of the proposed hybrid FS method in microarray data processing and classification versus some of the recently proposed FS methods in the literature.

Table 10
The proposed hybrid FS method versus some state-of-the-art hybrid FS methods using NB classifier based on accuracy and number of selected genes over CNS dataset.

Reference	Method	Accuracy	#Genes
Bolón-Canedo et al. (2015a)	DRFO-Cons with C4.5	80	11
Bolón-Canedo et al. (2015a)	DRF-IG 10% with SVM	65	2
Salem et al. (2017)	IG/SGA	86.67	38
Our Proposed methods			Features-list
	PHFS-IWD+NB	86.6	5 [3155,2461,2780,2615,1158]
	PHFS-IWDTS+NB	93.3	5 [26,1537,852,2218,1054]
	PHFS-IWDNLSA+NB	93.3	5 [2126,2859,1085,727,2160]
	PHFS-IWDHC+NB	86.6	5 [1801,1413,2248,421,250]

Table 11
The proposed hybrid FS method versus some state-of-the-art hybrid FS methods using NB classifier based on accuracy and number of genes over Ovarian Cancer dataset.

Reference	Method	Accuracy	#Genes
Bolón-Canedo et al. (2015a)	DRFO-INT with KNN	100	11
Bolón-Canedo et al. (2015a)	DRFO-CFS	100	16
Baliarsingh et al. (2020)	ANOVA-EJFOA-SVM	100	4
Our Proposed methods			Features-list
	PHFS-IWD+NB	96.87	2 [6282, 5029]
	PHFS-IWDTS+NB	98.43	2 [7373, 4763]
	PHFS-IWDNLSA+NB	100	2 [5910, 5707]
	PHFS-IWDHC+NB	98.43	2 [5910, 561]

From the comparative analysis of results, it is observed that the proposed methods improve accuracy with an average equal to 8.92% in three out of five datasets and decrease the number of genes with a percentage of 58.5% in all five datasets compared with some state-of-the-art methods.

On the other hand, we compared the proposed hybrid FS method with improved IWD using TS, NLSA, and HC local search algorithms named (PHFS-IWDTS, PHFS-IWDNLSA, and PHFS-IWDHC) respectively versus a hybrid method using an original IWD as a wrapper without any LS algorithms (PHFS-IWD) based on F-score, where PHFS-IWDNLSA, PHFS-IWDTS, and PHFS-IWDHC methods outperform PHFS-IWD method with an average of increment in F-score equal to 8.27%, 4.26%, and 2.67% respectively.

The PHFS-IWDNLSA method improves the F-score value with an average of 4.15% from PHFS-IWDTS, and 5.67% from PHFS-IWDHC, while PHFS-IWDTS outperformed PHFS-IWDHC with an average of 1.6%.

In the future, we will apply more local search algorithms and comparing the performance using different classifiers or applying these LS algorithms over other algorithms rather than IWD.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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